



Pain reduction of acupoint electrical stimulation for patients with spinal surgery: A placebo-controlled study

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ABSTRACT

Background: Acupoint electrical stimulation (AES) is commonly used for pain management. However, its true or placebo effect to achieve pain relief needs to be verified.

Objective: This study aimed to examine the true effect of AES to reduce postoperative pain in patients with spinal surgery receiving patient-controlled analgesia (PCA).

Method: A placebo- and sham-controlled study was conducted. Participants were randomly assigned to intervention with AES at true acupoints (the AES group, $n = 30$), AES at sham acupoints (the sham group, $n = 30$), or no intervention with AES (the control group, $n = 30$). Outcomes were assessed according to the amount of pain experienced and analgesics used.

Results: There were significant differences among the three groups in pain relief across time, and the occurrence of PCA button pushed and amount of analgesics used. The beneficial effects of AES were discernible when compared to the sham and the control.

Conclusions: AES at the true acupoints effectively reduced postoperative pain and analgesic usage. AES has now been implemented into healthcare and it is recommended that nurses be provided with the opportunity to earn their AES skills. More studies evaluating the effects of AES over a longer period and on pain after different surgical procedures are suggested.

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What is already known about this topic?

- The source of pain after neurosurgical procedures is the operative wound, and spinal cord surgery can result in more pain than anticipated.
- PCA is an effective and safe method for postoperative pain control in patients who undergo spinal surgery, but

50–60% was still suffering from moderate pain at first 24 h after starting of PCA.

- Evidence of AES-induced postoperative pain relief is needed; in particular its true effects need to be distinguished from its placebo effects.

What this study adds?

- AES, compared to placebo and no intervention, markedly decreased mild pain over a prolonged period.
- AES at true acupoints reduced postoperative pain and the need for analgesic (opiate) pain relief in patients with spinal surgery.
- Although the PCA morphine dosage required for analgesia is reduced by AES, the amount was obviously insufficient.

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1. Introduction

In the past two decades, 77% of patients have pain after chest, abdomen, or orthopedic surgery; 80–85% of these patients experience moderate to severe pain (Pellegrini et al., 1986; Warfield and Kahn, 1985). Neurosurgical procedures and wound pain caused more than the anticipated amount of pain, especially in patients with spinal surgery (Klimek et al., 2006). Unfortunately the pain of only 29.8% of the 63.8% of patients with spinal surgery with moderate wound pain was treated effectively (Lin et al., 2001).

Patient-controlled analgesia (PCA) is a popular method of pain management. Various opioids have been used for PCA, including morphine, meperidine, hydromorphone, and fentanyl, especially morphine which is by far the most commonly used opioid in this setting (Hutchison et al., 2006). PCA for postoperative pain management may thus be an effective and safe method of pain control producing high satisfaction in patients (Gepstein et al., 2007). However, almost one half of the patients using PCA complain of nausea and vomiting due to morphine side effects (Chung et al., 2007). More importantly, patients with fewer side effects had more satisfaction.

Acupoint electrical stimulation (AES), like acupuncture (an established technique of traditional Chinese medicine [TCM]), is commonly used for pain management. It relies on the concept that particular acupoints can be stimulated with electrical current to release endorphins within the central nervous system (Johnson, 2001). AES-induced pain reduction has been reported. For instance, AES significantly reduced osteoarthritic knee pain (Leung, 2003), and pain of extracorporeal shock-wave lithotripsy (Resim, 2005). AES had sedative and analgesic effects during artificial abortion operation and decreased the adverse reactions (Cheng et al., 2010). Moreover, combining pain control or analgesia with electroacupuncture decreased the dose of opiate and delayed the first opiate request (Wei, 2001), reduced the dosage of 17% propofol and 14% fentanyl in the operation (He and Yang, 2008), and provided an adjuvant effect to enhanced the analgesic effect in propofol-fentanyl anesthesia (Si et al., 2009).

Some studies argued that the effects of acupoint stimulation were placebo effects. In Chen et al.'s study (2005), PCA plus AES (but not control and sham treatment) reduced the amount of morphine used and increased the amount of postoperative pain relief. Similarly, Kober et al. (2002) found that true acupoint acupressure (but not sham and control treatment) reduced trauma pain. Chao et al. (2007) found that electroacupuncture, compared to a placebo, relieved labor pain. However, Lin et al. (2002) found that both electroacupuncture and sham treatment reduced postoperative PCA morphine use, and sham treatment relieved pain without side effects. Currently, chemical analgesia is the main strategy for clinical relief of postoperative pain. Notably, morphine has side effects and is associated with higher medical costs. AES could be useful as a supplementary analgesic, but it has not been well studied (Kotzé and Simpson, 2007). In addition, whether the relief of pain by AES is a true effect or placebo effect has not been verified.

2. Purpose statements

This study aimed to examine the true effects of AES on reducing pain intensity, pain influence, the demand for PCA, and opiate dose used during the first 24 h after spinal surgery. First, significant differences in the change of pain reduction across time by the different interventions were hypothesized. Second, significant differences in pain intensity and influence, the demand for PCA and opiate dose were hypothesized among groups.

3. Methods

3.1. Research design

This study was a single-blind, randomized, placebo-controlled trial. The randomization was achieved through a computer-generated numbers and information on intervention allocation sequential without sealed envelopes. Individual subjects and medical staff did not know in which group the subjects were.

3.2. Participants and setting

The study sample consisted of patients who were scheduled for surgery to correct non-traumatic lumbar spine injuries at a 3000-bed medical center in northern Taiwan. Inclusion criteria were: age 18 years and older, American Society of Anesthesiologists physical status I and II, general anesthesia, and consent to PCA with morphine. After the end of the operation, the intravenous PCA was connected to the central venous line. Exclusion criteria were: usage of a pacemaker, history of arrhythmia or epilepsy, opiate dependence, clinically significant cardiovascular disease, and cutaneous lesions at the application sites. To detect a medium effect size ($f=0.3$) at 5% level of significance with 80% power using repeated-measures, 72 subjects were minimally required (Cohen, 1988). With an estimated follow-up loss, 99 overall were thought to be necessary. Fig. 1 shows the flow chart of participants of this study. Initially, 99 patients were eligible, and then 90 who signed the informed consent were included in the study and randomly assigned to one of the three groups: the AES group ($n=30$) received AES at the true acupoints; the sham group ($n=30$) received AES in the same manner but at sham acupoints; and the control group ($n=30$) received no AES intervention. By the end of the study, data from a total 90 participants were used to analyze.

3.3. Intervention

An electrical device stimulating the acupoints of Weizhong (BL40), Yanglingquan (GB34), Shenmen (HT7), and Neiguan (P6) was used for AES (Fig. 2). The two pairs of rubber electrodes were placed at the acupoints, which were stimulated by alternating pulses of a 2 Hz/100 Hz dense-disperse waveform and 0.25 min pulse duration. A current output (4–7 mA) (Han, 2004) was adjusted and depended on individual body weight. Participants received AES in a supine position and 4 acupoints simultaneously stimulated for 20 min: the first in the third postoperative

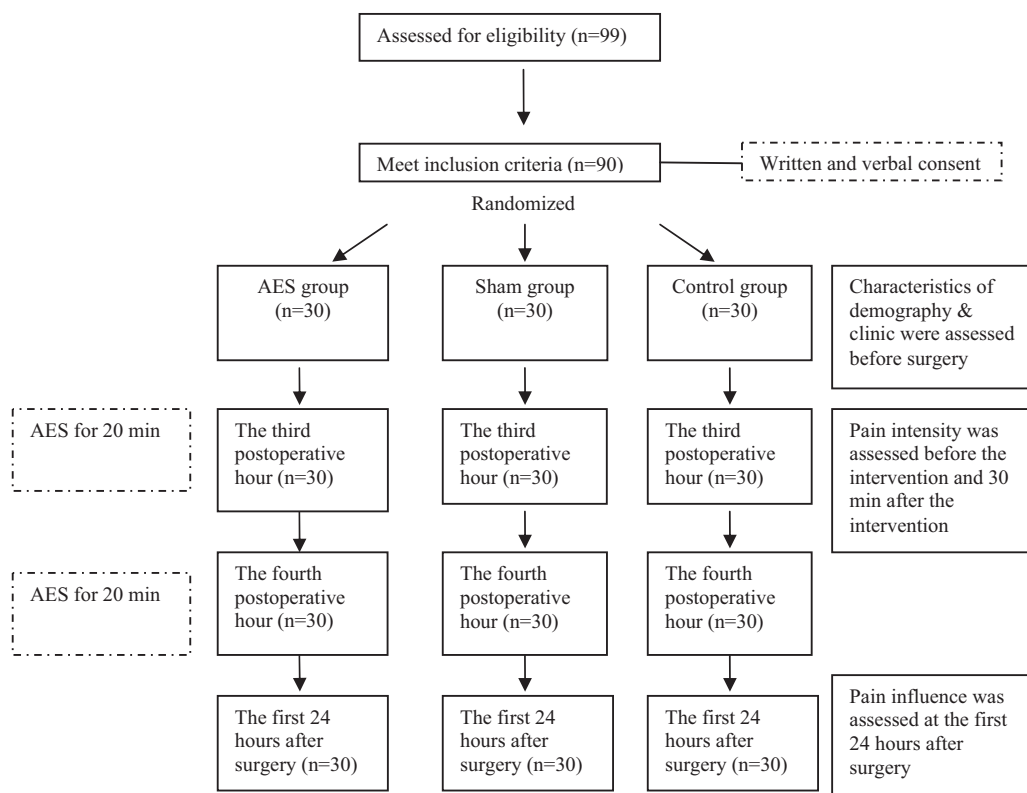


Fig. 1. The flow of research design, participants' recruitment, intervention and assessment.

hour (calculated from the time a patient was moved to the recovery room) and the second in the fourth postoperative hour. In the same manner, the sham group received AES at points 2 cm from the true acupoints, not meridian acupoints. The AES interventions and data collection were performed by the researchers. Two TCM physicians verified the conduct of the AES, which consisted of

selection of acupoints, current frequency and output, waveform, and length of stimulation.

3.4. Measures

Demographic characteristics included gender, age, body weight, and body height. Clinical characteristics included disease diagnosis, the worst preoperative pain, the number of lumbar surgeries, pain intensity, types of surgery, operation duration, and amounts of blood loss. The primary outcomes involved pain severity and pain interference. The Brief Pain Inventory (BPI), including the subscales of pain intensity and its interference, was used to measure the severity of postoperative pain and impact of pain on daily functions (Ger et al., 1999). The pain intensity referred to the level of present pain and the level of the worst and average pain in the last 24 h. The impact of postoperative pain on daily functions included the impact on ability to perform daily activities (e.g., eating, dressing, and toilet use), bodily functions (e.g., deep breathing and coughing), physical mobility, talking, sleep, and emotion. A simple numeric rating scale was used, with 0 indicating no pain or no interference and 10 indicating worst pain possible or interferes completely. Moreover, 0 represented no pain at all, 1–3 for mild pain, 4–6 for moderate pain, 7–9 for severe pain, and 10 for intolerable pain. In this study Cronbach's alpha, an internal consistency test which measures the degree of which the items consistently measure the underlying latent construct, was 0.94 and 0.87 for the pain severity and pain interference, respectively. The

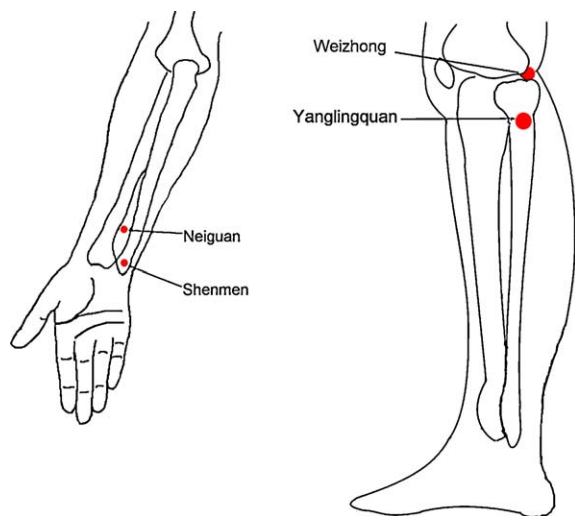


Fig. 2. Diagram of acupoints location, include Shenmen, Neiguan, Weizhong, and Yanglingquan.

secondary outcome was opiate requirement, including time to initial request for PCA analgesia, number of times the button was pushed to receive analgesic treatment, and the total amount of opiate used during the first 24 h after surgery.

3.5. Procedures and data analysis

First, ethical approval was obtained from the Institutional Review Board of the study hospital. Informed consent was signed by all participants. Second, all participants were given the same regimen of preoperative medication, which was general anesthesia with intravenous thiopental (2.1 mg/kg) and fentanyl (4.3 µg/kg), and connected to a PCA device programmed to deliver a patient-triggered 1 mg bolus of morphine with a 10 min lockout interval between doses. Last, the outcome measures were administered at baseline prior to the intervention. Pain was assessed 30 min after the first and second AES intervention (posttest 1 and 2, respectively), whereas BPI data and opiate requirement were collected at 24 h after surgery. In addition, adverse effects of AES were recorded.

Statistical analysis was performed using SPSS 18.0 version. The 5% level of significance was used to confirm. Descriptive statistical analysis was used to examine differences in demographic and clinical characteristics; inferential statistical analysis consisted of the chi-square test, the one-way analysis of variance with post hoc tests, and a general linear mixed model with fixed effects of intervention, time, and the interaction of these two, and a random subject effect, followed by a Bonferroni test to compare the effects of pain reduction.

4. Results

4.1. A comparison of differences in demographic and clinical characteristics at baseline

The participants were divided into three groups, with 30 in each group in the study. Table 1 summarizes the

demographic and clinical characteristics of each group. No significant differences were found among the AES group, sham group, and control groups in gender, age, body weight, body height, other chronic diseases, diagnosis on admission, number of lumbar surgeries, the worst preoperative pain, the average preoperative pain, operation duration, and amount of blood loss ($p > 0.05$).

4.2. The results of AES on pain intensity

Fig. 3 shows VAS scores across time for each group. Before the intervention, the baseline VAS scores were similar among groups (mean, 4.37 ± 1.78 in the AES, 5.43 ± 2.12 in the sham, and 4.91 ± 1.89 in the control; $F(2, 87) = 2.47$, $p = 0.09$). The result of the general linear mixed model on the main effect of group was significant difference in the change of pain reduction across time in terms of the VAS scores ($F(2, 87) = 6.80$, $p = 0.002$); pain was significantly lower in the AES group than the control ($p < 0.007$) or sham group ($p = 0.005$), but not significantly different between the sham and the control ($p = 1.00$). Regardless of interventions, the statistical significance in the main effect of time was found ($F(2, 174) = 147.83$, $p < 0.001$). The interaction between group and time was also significant ($F(2, 174) = 6.04$, $p < 0.001$).

4.3. The results of AES on BPI and opiate requirement

Table 2 summarizes the effect of AES intervention on pain intensity and its influence in various activities. There was a significant between-group difference in the worst ($F(2,87) = 4.86$, $p = 0.01$) and average ($F(2, 87) = 4.71$, $p = 0.01$) pain intensity 24 h after surgery. Scheffe's post hoc analysis confirmed the significance of the difference between the AES and the sham ($p = 0.03$) or the control ($p = 0.03$) for the worst and the average pain intensity. There were significant between-group differences in the ability of pain to interfere with change of position ($F(2,$

Table 1
Homogeneity of demographic and medical characteristics at baseline.

Variables	AES	Sham	Control	$F(df)$ or $\chi^2(df)$ (p)
Gender (n/%)				$\chi^2(2) = 2.93$ (0.23)
Male	10/33.3	9/30.0	15/50.0	
Female	20/66.7	21/70.0	15/50.0	
Age ($M \pm SD$)	60.7 ± 12.0	57.2 ± 15.7	63.2 ± 14.0	$F(2, 87) = 1.34$ (0.27)
Body weight ($M \pm SD$)	158.4 ± 7.4	157.1 ± 9.9	160.3 ± 9.1	$F(2, 87) = 1.00$ (0.37)
Body height ($M \pm SD$)	65.4 ± 10.4	63.9 ± 13.6	64.5 ± 13.3	$F(2, 87) = 0.11$ (0.90)
Other chronic disease (n/%)				$\chi^2(6) = 11.63$ (0.07)
None	20/66.7	17/56.7	11/36.7	
One	4/13.3	8/26.7	9/30.0	
Two	5/16.7	5/16.6	5/16.7	
Three	1/3.3	0	5/16.7	
Admitted diagnosis (n/%)				$\chi^2(4) = 6.63$ (0.16)
HIVD	7/23.3	5/16.7	2/6.7	
Spondylolisthesis	10/33.3	10/33.3	6/20.0	
Lumbar spinal stenosis	13/43.4	15/50.0	22/73.3	
Number of lumbar surgery ($M \pm SD$)	0.4 ± 0.9	0.3 ± 0.8	0.2 ± 0.5	$F(2, 87) = 0.55$ (0.58)
The worst preoperative pain ($M \pm SD$)	7.2 ± 1.9	7.2 ± 1.1	7.2 ± 1.6	$F(2, 87) = 0.01$ (0.99)
The average preoperative pain ($M \pm SD$)	1.5 ± 1.1	1.3 ± 1.0	0.9 ± 0.9	$F(2, 87) = 2.56$ (0.08)
Operation duration (min) ($M \pm SD$)	217.5 ± 98.1	267.6 ± 125.5	267.6 ± 125.5	$F(2, 87) = 1.70$ (0.19)
Amount of blood loss (cc) ($M \pm SD$)	450.8 ± 453.8	477.7 ± 459.1	433.7 ± 339.0	$F(2, 87) = 0.08$ (0.92)

$n = 30$ in each group; F -test for numerical data or chi-square test for categorical data.

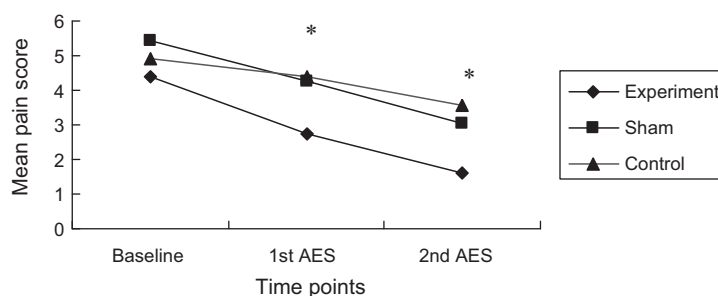


Fig. 3. Pain reduction across time for each group, data were presented as mean of VAS (0–10); * pain scores were significantly lower in the experimental group than sham and control groups, F test < 0.05.

Table 2
Comparisons of pain intensity and pain influence among the three groups.

Variables	AES	Sham	Control	$F(df)$ (p)
	$M \pm SD$	$M \pm SD$	$M \pm SD$	
<i>Pain intensity</i>				
The worst pain	5.4 ± 1.6	6.5 ± 1.6	6.5 ± 1.7	$F(2, 87) = 4.86$ (0.01), #S > A, C > A
The average pain	2.1 ± 1.3	3.0 ± 1.4	3.0 ± 1.4	$F(2, 87) = 4.71$ (0.01), #S > A, C > A
<i>Pain influence</i>				
Special activities	1.6 ± 1.8	1.9 ± 2.1	2.3 ± 2.5	$F(2, 87) = 0.66$ (0.52)
Change of position	5.2 ± 1.5	6.1 ± 1.5	6.4 ± 1.7	$F(2, 87) = 4.79$ (0.01), #C > A
Daily activities	5.3 ± 1.4	6.1 ± 1.5	6.5 ± 1.7	$F(2, 87) = 4.81$ (0.01), #C > A
Talking	0.3 ± 0.7	0.8 ± 1.3	1.1 ± 1.3	$F(2, 87) = 3.62$ (0.03), #C > A
Sleep	1.1 ± 1.2	2.1 ± 2.2	2.4 ± 1.7	$F(2, 87) = 4.76$ (0.01), #C > A
Emotion	0.2 ± 0.6	0.3 ± 0.8	0.4 ± 0.8	$F(2, 87) = 1.01$ (0.52)

$n = 30$ in each group; #: Scheffe post hoc test; A: the AES; S: the sham; C: the control.

Table 3
Comparisons on opiate demands and doses among the groups.

Variables	AES	Sham	Control	$F(df)$ (p)
	$M \pm SD$	$M \pm SD$	$M \pm SD$	
The timing of the first request for PCA morphine (min)	98.5 ± 160.1	92.8 ± 109.1	83.4 ± 133.6	$F(2, 84) = 0.09$ (0.92)
Occurrence of button pushed (time)	24.9 ± 28.5	47.4 ± 39.7	76.9 ± 86.1	$F(2, 84) = 6.05$ (0.004), #C > A
24 h opiate dose (mg)	19.3 ± 9.7	21.6 ± 13.1	28.0 ± 12.1	$F(2, 84) = 4.26$ (0.017), #C > A

$n = 30$ in each group; #: Scheffe post hoc test; A: the AES; S: the sham; C: the control.

87) = 4.79, $p = 0.01$), daily activities ($F(2, 87) = 4.81$, $p = 0.01$), talking ($F(2, 87) = 3.62$, $p = 0.03$), and sleep ($F(2, 87) = 4.76$, $p = 0.01$), but not bodily functions ($F(2, 87) = 0.66$, $p = 0.52$) and emotion ($F(2, 87) = 1.01$, $p = 0.52$).

Table 3 shows between-group differences in the number of times the patients pushed buttons to achieve pain relief ($F(2, 84) = 6.05$, $p = 0.004$) and the amount of opiate required during the first 24 postoperative hours ($F(2, 84) = 4.26$, $p = 0.017$) but not in the timing of the first request for PCA morphine ($F(2, 84) = 0.09$, $p = 0.92$). Scheffe's post hoc analysis confirmed the significance of the difference between the AES and the control in the number of times the patients pushed buttons ($p = 0.004$) and the amount of opiate required ($p = 0.023$). The AES groups had not adverse effects associated with AES intervention during the present study period.

5. Discussion

As shown in Fig. 3, pain in all groups decreased gradually over time. However, the participants in the AES group decreased markedly from moderate to mild pain, and their progression of pain reduction was greater than those in the sham or control group. This result was similar to other studies. Chen et al. (2005) found that pain due to total knee replacement decreased at various time points after treatment with electroacupuncture plus PCA. Lin et al. (2002) found that pain due to abdominal hysterectomy was alleviated by electroacupuncture. Similarly, Lin (2004) used electroacupuncture or acupuncture in conjunction with PCA to alleviate pain after Caesarean section. Moreover, it is noted that the second AES intervention reduced pain from moderate to mild, while the decrease in

response to sham and control intervention was less obvious and less marked. AES therefore could maintain this decrease in pain intensity. In addition, the findings support Melzack (1975) that the true effect of AES is significantly more effective than placebo contributions.

As shown in Table 2, the worst and average pain intensities during the first 24 h after surgery were lower in the AES group than the other two groups, indicating that AES interventions alleviate pain rapidly and persistently. Unlike Gepstein et al. (2007) who demonstrated a reduction in average pain intensity from severe to moderate by PCA in 24 h after spinal surgery, this study showed that PCA plus AES was able to reduce average pain intensity from moderate to mild. In our previous study pain control multimedia education with PCA used was also able to lower average pain intensity to mild-moderate (which was even lower than that in the control group) in the first 24 h after spinal surgery (Yeh et al., 2007). Although PCA plus AES alleviated pain for lumbar surgery, the pain exhibited at a certain level and to obviously impact daily activities within first 24 h after surgery.

From Table 3, the PCA morphine dosage required for analgesia was reduced by AES interventions. This is in agreement with Hamza et al. (1999) that treated the abdominal pain from gynecologic surgery with 2 Hz, 100 Hz, and 2/100 Hz alternating frequency electrical stimulation, and found that all three protocols reduced the need for opioid analgesia and the last was most effective. A mixed 2 and 100 Hz frequency stimulation facilitates central nervous system release of enkephalin, β -endorphin, endomorphin, and dynorphin to maximize the analgesic effect (Han, 2004). The peak effect occurs at 20 min after stimulation (Anderson and Holmgren, 1975). It is noted that the amount of analgesic requirement was obviously less in the present study, in contrast to the amount used in the study by Aveline et al. (2006). Cultural and social influences can affect the demand for analgesics and therefore Chinese patients request fewer analgesics to reduce postoperative pain than American patients. Other studies (Lin, 2000; Streltzer and Wade, 1981) have come to a similar conclusion that Chinese surgical patients request and receive less amount of analgesic medication. Patients may consider the possibility of drug addiction (Cekmen et al., 2007; Yeh et al., 2007) and do not want to be stigmatized as a pain complainer (Lin, 2000). Patients in the present study therefore may have hesitated and failed to push the PCA button to achieve pain relief. They endured their moderate pain and remained active.

Acupoint stimulation is a common treatment for reducing pain in the Chinese population. AES is applied in a non-invasive and easy-to-use fashion. This study demonstrated less pain reduction and greater number of opiate doses administered in sham-treated and untreated control patients. Notably, treatment expectations, prior beliefs, appearance of the therapy, and interactions with therapists or research staff seem to contribute to the placebo effect (Wager et al., 2004). The patient's expectation is the major contributor to increasing treatment efficacy (Montgomery and Kirsch, 1997). A neuroimaging study showed that placebo acupuncture produced detectable analgesia after enhancement of expectancy to

acupuncture analgesia (Kong et al., 2006). This study was however randomized and placebo-controlled, and participants as well as medical staff were blinded to the procedure assignment; thus, expectations in both the true and sham AES were the same. The results of this study were also consistent with those of other double-blind randomized placebo-controlled trials (Kober et al., 2002; Lin et al., 2002). Moreover, use of sham and control groups can provide the means to distinguish clearly between true and placebo effects. As previous mentioned the true effect of AES was more effective than placebo contributions in this study. Placebo analgesia is related to decreased brain activity, and thereby to a change in pain perception (Bingel et al., 2006; Napadow et al., 2006). However, the clinical effects of multiple acupuncture analgesia treatments on the brain are cumulative and extend beyond the length of each session (Price et al., 1984; Carlsson, 2002). In contrast, pain reduction is immediate after placebo acupuncture but declines soon thereafter (Wang and Tronnie, 2000). Short-term acupuncture and placebo treatments affect similar networks, but the effects of acupuncture on specific brain areas are prolonged by treatment repetition (Dhond et al., 2007).

5.1. Limitations

This study was restricted by experimental assumptions, including that postoperative pain is strongest a few hours after narcotic dissipation. The outcome was assessed at only 4 h and 24 h after surgery. Extension of monitoring of pain intensity, level of interference with activity, frequency and amount of analgesic use, morphine-induced side effects, and so on to 3 postoperative days would have provided a more complete assessment of AES in postoperative pain management. The lack of funding prevented us from assessing biochemical outcomes such as enkephalin, β -endorphin, or endomorphin. Besides, the possibility of the threat to internal validity due to a single-blind design may not be avoided. The findings were from only one hospital and may not be generalizable to other hospitals and settings. Future investigations should include multiple settings.

6. Conclusions and implications

This study demonstrated that AES at the true acupoints reduces pain across time after spinal surgery and decreases the need for opiate doses administered through PCA. AES is a simple, convenient, noninvasive, and economical way to control pain. AES has now been implemented into healthcare and it is recommended that nurses be provided with the opportunity to earn their AES skills. This study encourages nurses to learn and implement this simple technique in their routine clinical care. Through integration of traditional Chinese and Western remedies of pain management, health professionals can improve the quality of clinical care. The effect of AES to relieve postoperative pain after different types of surgery should be tested. Moreover PCA pain treatment is associated with morphine-induced side effects, which directly influence the number of times the button is pushed to receive analgesic relief. Therefore, further study is needed to optimize pain

management using PCA morphine plus AES and to minimize morphine-induced side effects.

Ethical approval

Ethical approval was obtained from the Institute of Review Board in the study hospital; written informed consent was signed by and obtained from all participants. All participants were provided written and verbal information about the study, and made aware that data collected by the researcher were remained confidential; they would be free to withdraw.

Funding

None.

Conflict of interest

None of the authors has any potential conflicting interest in this study.

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